

- 7 (a) Fig. 7.1 shows part of the process of respiration in a mitochondrion.

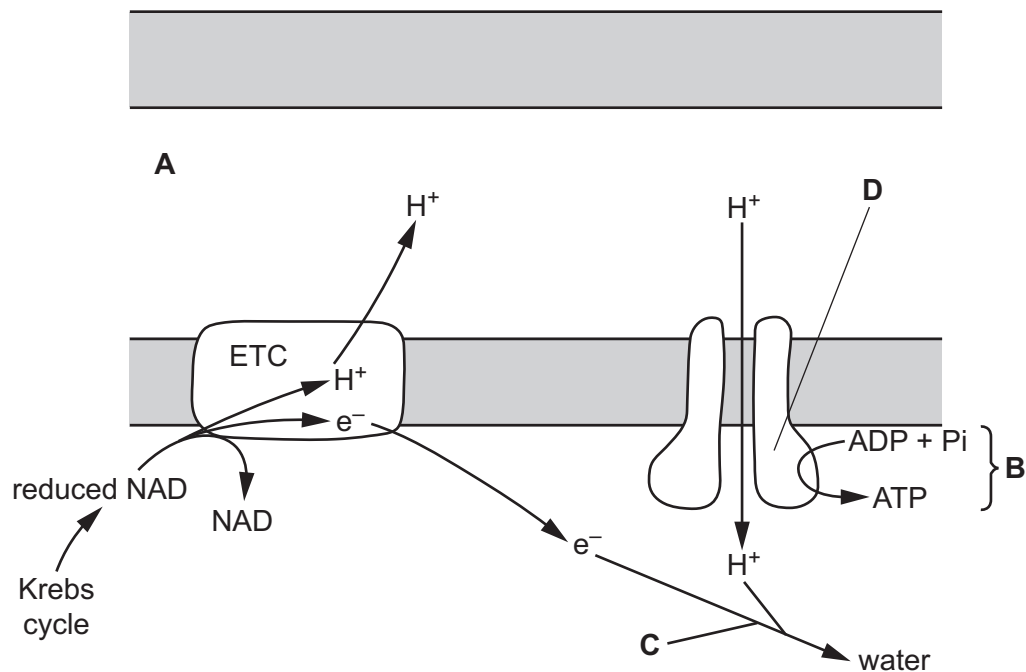


Fig. 7.1

- (i) With reference to Fig. 7.1, name:

area **A**

process **B**

substance **C** [3]

- (ii) State the type of chemical compound that is represented by **D**.

..... [1]

- (iii) State the process by which ATP can be synthesised directly during glycolysis or the Krebs cycle.

..... [1]

The elephant seal, *Mirounga angustirostris*, spends most of its life in the ocean.

Fig. 7.2 is an elephant seal.



Fig. 7.2

- (b) Elephant seals can stay underwater for up to two hours. During this time, respiration continues.

Fig. 7.3 shows the mass of blood in the body, as a percentage of total body mass, for the elephant seal and for three other mammals.

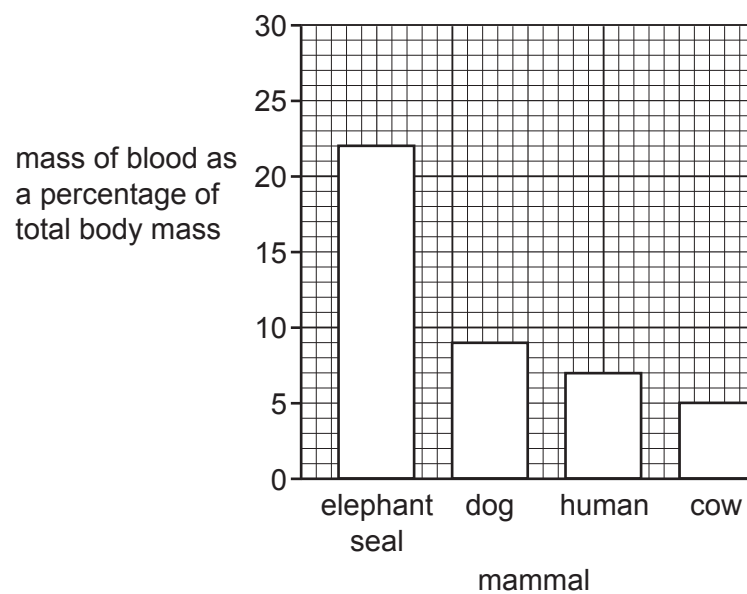


Fig. 7.3

- (i) Fig. 7.3 shows that elephant seals have a higher mass of blood as a percentage of total body mass than humans.

Calculate how many times greater this figure is for elephant seals compared to humans.

Show your working and write your answer to **two** decimal places.

answer [2]

- (ii) Suggest why an elephant seal needs such a large mass of blood as a percentage of total body mass.

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..... [2]

- (c) Elephant seals have a very thick layer of adipose tissue under their skin. Adipose cells are rich in fat molecules.

Suggest why 1 g of fat will produce more ATP than 1 g of carbohydrate as a respiratory substrate in aerobic respiration.

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..... [2]

- (d) (i) Adipose tissue plays an important role in thermoregulation in elephant seals.

State the role of adipose tissue in thermoregulation.

..... [1]

- (ii) The hypothalamus in the brain is the control centre for thermoregulation.

Outline how a change in temperature of the external environment results in an impulse arriving at the hypothalamus.

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..... [2]

- (iii) When the blood temperature of a mammal decreases, one response is that its body secretes more adrenaline.

Suggest how an increase in adrenaline results in an increase in blood temperature.

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..... [2]

[Total: 16]